Real-time wearable platform: haptic delivery on Apple Watch

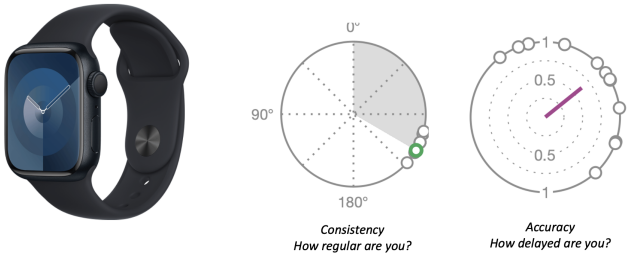
Visiting Researcher • Oticon Medical / Eriksholm innovation team, Copenhagen • 2024 • Swift, OSC

Linked: code repository - github.com/VillejoubertLouis

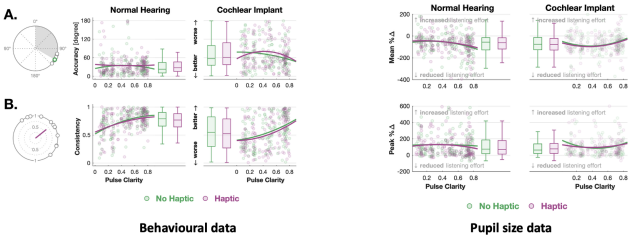
Can an Apple Watch improve speech understanding and music sound quality through haptic feedback?

What I built

I built the full pipeline end to end: a Swift iPhone/Apple Watch application linking the watch to a laptop over a network protocol (OSC) to deliver controlled haptic feedback in real time, synchronised to the audio. Evaluation combined structured user interviews with quantitative measures (pupillometry, eye movements, sound-quality assessment and rhythm reproduction), and I captured movement and response data from the watch in reaction to the haptic cues.



The two behavioural measures collected during the experiment.



Effect of haptic feedback on behaviour (left) and listening effort (right).